SlipSurfaceIQ LE

# Python Scripting Tutorial

Build and edit complete slope models with Python

Version 2026.6.1

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# 01 Getting started

Python scripting builds or edits a complete model in a few lines: geometry, materials, water, loads, supports, settings and notes.

Open the editor with **Tools → Python scripting**. It opens empty. Write a script, or open a **.py** file, then press **Run / Apply**. **Save .py** stores the editor text; **Tutorial** opens this document.

## How a script runs

• The script receives **project**, a *copy* of the open model. Change it in place, or assign a new project to the same name.

• When the script finishes, the result is validated and its geometry is built. If everything is valid, all changes are applied as **one undoable step** (Ctrl+Z in the Modeler reverts the whole script).

• If the script raises an error or the model is invalid, the open project is left unchanged and the message appears in the output pane.

• Anything printed with **print()** is shown in the output pane.

• By default the editor runs scripts in **restricted mode**: only **import math** and **from slipsurface.project / loads / materials / geometry / lem / analysis import …** are accepted, and files, **exec**/**eval**, names starting with an underscore and **project.save()**/**load()** are refused before anything runs. Every example in this tutorial runs in restricted mode.

• Tick **Full Python access** to run ordinary Python (any import, files). The editor asks for confirmation first. Restricted mode guards against scripts received from others; it is not a substitute for reading a script before you run it.

**Units**
Lengths in m, forces in kN, stresses in kPa, unit weights in kN/m$^3$, angles in degrees. The y axis points up.

**Trusted code only**
Scripts are ordinary Python with your account's permissions. A script opened from a file asks for confirmation before its first run. Projects never run scripts on their own.

## Your first script

first_script.py

```
from slipsurface.project import empty_project, MaterialSpec

project = empty_project()
project.summary.title = "My first slope"
project.external = [[0, -10], [60, -10], [60, 10], [40, 10], [20, 0], [0, 0]]
project.materials = [MaterialSpec(name="Clay", gamma=19, c=8, phi=26)]
project.methods.methods = ["bishop", "spencer"]
print("Materials:", project.material_names())
```

Press **Run / Apply**; the slope appears in the Modeler and is ready for **Compute**.

# 02 Names available in a script

| Name | What it is |
|---|---|
| `project` | Copy of the open model (Project). The result is what it holds at the end. |
| `empty_project()` | New project with one default material 'Soil 1' and no geometry. |
| `Project` | The project class; Project() has no materials. |
| `MaterialSpec(...)` | A material: name, model and parameters. |
| `SupportType(...)` | A support definition (anchor, nail, geosynthetic, …). |
| `FacingType(...)` | A shotcrete + mesh facing definition. |
| `print(...)` | Write to the output pane. |

Everything else in Python works as usual: loops, functions, the **math** module and imports from the application, e.g. `from slipsurface.loads import SUPPORT_KINDS`.

## Useful project helpers

| Call | Returns |
|---|---|
| `project.name` | The title (same as project.summary.title). |
| `project.material_names()` | List of material names. |
| `project.material(name)` | The MaterialSpec with that name, or None. |
| `project.support_type(name)` | The SupportType with that name, or None. |
| `project.facing_type(name)` | The FacingType with that name, or None. |
| `project.validate()` | Raise ValueError if the model is inconsistent. |
| `project.copy()` | An independent deep copy. |

# 03 Project summary

Shown in the window title and on the report cover.

| Field | Default | Meaning |
|---|---|---|
| `title` | `'New Project'` | Project title (window caption, report cover). |
| `analysis` | `''` | Analysis description. |
| `author` | `''` | Author shown in the report. |
| `company` | `''` | Company shown in the report. |
| `date_created` | `today` | Creation date, YYYY-MM-DD. |
| `comments` | `['', '', '', '', '']` | Five free comment lines. |

summary.py

```
project.summary.title = "Highway cut, km 12+400"
project.summary.author = "H. Engineer"
project.summary.company = "Geotech Ltd."
project.summary.comments[0] = "Long-term drained analysis"
```

# 04 Geometry

Points are [x, y] lists in metres. The external boundary is one closed polygon; material boundaries split it into regions.

| Field | Format | Meaning |
|---|---|---|
| `external` | `[[x, y], ...]` | External boundary polygon, at least 3 points (closed automatically). |
| `material_boundaries` | `[[[x, y], ...], ...]` | Polylines that divide the domain into regions. |
| `assignments` | `[{'point': [x, y], 'material': name}]` | A point inside a region gives it a material. Unassigned regions use the first material. |

## A layered slope

layers.py

```
project.external = [[0, -12], [70, -12], [70, 12], [45, 12], [25, 0], [0, 0]]
project.materials = [
    MaterialSpec(name="Fill", gamma=19, c=5, phi=32, color="#f0c800"),
    MaterialSpec(name="Clay", gamma=18, c=12, phi=22, color="#3cb371"),
]
project.material_boundaries = [[[0, -4], [70, -4]]]
project.assignments = [
    {"point": [50, 5], "material": "Fill"},
    {"point": [50, -8], "material": "Clay"},
]
```

**Build boundaries with loops**
Geometry is plain data, so loops and functions work well — for example, several horizontal layers from a list of elevations.

layers_loop.py

```
tops = [-3, -7]                      # layer boundary elevations
names = ["Sand", "Silt", "Clay"]
project.materials = [MaterialSpec(name=n, c=5 + 5 * i, phi=32 - 4 * i)
                     for i, n in enumerate(names)]
project.material_boundaries = [[[0, y], [60, y]] for y in tops]
mids = [-1.5, -5, -9]                # a point inside each layer
project.assignments = [{"point": [50, y], "material": n}
                       for y, n in zip(mids, names)]
```

# 05 Materials

Every material has common fields and the parameters of its strength model.

| Field | Default | Meaning |
|---|---|---|
| `name` | `'Soil 1'` | Unique material name (assignments refer to it). |
| `model` | `'MC'` | Strength model key, see the models table. |
| `gamma` | `19.0` | Unit weight above the water table (kN/m3). |
| `gamma_sat` | `20.0` | Saturated unit weight (kN/m3), used when use_sat is True. |
| `use_sat` | `False` | Use gamma_sat below the water table. |
| `use_water` | `True` | The water table acts on this material. |
| `hu` | `1.0` | Pore-pressure factor Hu applied to the water table head. |
| `color` | `'#f0c800'` | Fill colour, e.g. '#f0c800'. |

## Strength models

Set **model** to one of the keys below and give the parameters listed for it; unused parameters are ignored.

| model | Strength model | Parameters |
|---|---|---|
| `MC` | Mohr-Coulomb | `c, phi` |
| `UD` | Undrained (Tresca) | `su` |
| `UDG` | Undrained – linear with elevation | `su_top, su_bottom, su_y_top, su_y_bottom` |
| `SH` | SHANSEP | `shansep_s, shansep_m, ocr, shansep_su_min` |
| `HS` | Mohr-Coulomb (HS envelope) | `c, phi, hs_psi, hs_e50, hs_eoed, hs_eur, hs_m, hs_pref, hs_rf` |
| `SS` | Approximate peak ϕ (SANISAND) | `ss_m, ss_e, ss_ec0, ss_lambda, ss_xi, ss_nb, ss_pat` |
| `PL` | Power-law envelope | `coefficient, exponent, reference_pressure` |
| `PC` | Power Curve | `pc_a, pc_b, pc_c, pc_d` |
| `BB` | Barton-Bandis | `bb_phir, bb_jrc, bb_jcs, bb_l0, bb_ln` |

| model | Strength model | Parameters |
|---|---|---|
| `CN` | CNI Rock Mass | `cni_sigma_c, cni_a, cni_n, cni_t` |
| `HB` | Generalized Hoek-Brown | `sigma_ci, gsi, mi, d` |
| `HBD` | Hoek-Brown (mb, s, a) | `sigma_ci, hb_mb, hb_s, hb_a` |

## Mohr-Coulomb (MC)

τ = c′ + σ′n · tan ϕ′

| Parameter | Default | Meaning |
|---|---|---|
| `c` | `5.0` | Cohesion c (kPa) |
| `phi` | `30.0` | Friction angle ϕ (°) |

## Undrained (Tresca) (UD)

τ = su (total stress; pore pressure not used)

| Parameter | Default | Meaning |
|---|---|---|
| `su` | `25.0` | Undrained shear strength su (kPa) |

## Undrained – linear with elevation (UDG)

τ = su(y), linear from layer top to bottom (total stress)

| Parameter | Default | Meaning |
|---|---|---|
| `su_top` | `25.0` | Undrained strength at the top elevation (kPa) |
| `su_bottom` | `25.0` | Undrained strength at the bottom elevation (kPa) |
| `su_y_top` | `0.0` | Top elevation (m) |
| `su_y_bottom` | `-1.0` | Bottom elevation (m) |

## SHANSEP (SH)

τ = su = A + S · σ′v · OCR^m

| Parameter | Default | Meaning |
|---|---|---|
| `shansep_s` | `0.22` | S (normally consolidated su/σ′v) |
| `shansep_m` | `0.8` | m (exponent) |
| `ocr` | `1.0` | OCR (overconsolidation ratio) |
| `shansep_su_min` | `0.0` | A — minimum undrained strength (kPa) |

## Mohr-Coulomb (HS envelope) (HS)

τ = c′ + σ′n · tan ϕ′ (Mohr-Coulomb failure surface)

| Parameter | Default | Meaning |
|---|---|---|
| `c` | `5.0` | Cohesion c′ (kPa) |
| `phi` | `30.0` | Friction angle ϕ′ (°) |

| Parameter | Default | Meaning |
|---|---|---|
| `hs_psi` | `0.0` | Dilatancy angle ψ (°) |
| `hs_e50` | `30000.0` | E50,ref (kPa) |
| `hs_eoed` | `30000.0` | Eoed,ref (kPa) |
| `hs_eur` | `90000.0` | Eur,ref (kPa) |
| `hs_m` | `0.5` | m (stress dependency exponent) |
| `hs_pref` | `100.0` | pref (kPa) |
| `hs_rf` | `0.9` | Rf (failure ratio) |

## Approximate peak ϕ (SANISAND) (SS)

τ = σ′n tan ϕ(ψ), sin ϕ = 3Mb/(6+Mb), Mb = M·exp(−nb ψ)

| Parameter | Default | Meaning |
|---|---|---|
| `ss_m` | `1.25` | M (critical state stress ratio) |
| `ss_e` | `0.75` | Void ratio e |
| `ss_ec0` | `0.934` | ec0 (CSL reference void ratio) |
| `ss_lambda` | `0.019` | λc |
| `ss_xi` | `0.7` | ξ |
| `ss_nb` | `1.1` | nb |
| `ss_pat` | `101.3` | pat (kPa) |

## Power-law envelope (PL)

τ = a · pa · (1 + σ′n/pa)^b

| Parameter | Default | Meaning |
|---|---|---|
| `coefficient` | `0.2` | a |
| `exponent` | `0.5` | b |
| `reference_pressure` | `100.0` | pa (kPa) |

## Power Curve (PC)

τ = a · (σ′n + d)^b + c

| Parameter | Default | Meaning |
|---|---|---|
| `pc_a` | `1.5` | a |
| `pc_b` | `0.8` | b |
| `pc_c` | `0.0` | c (kPa) |
| `pc_d` | `0.0` | d (kPa) |

## Barton-Bandis (BB)

τ = σ′n · tan(ϕr + JRCn · log10(JCSn/σ′n))

| Parameter | Default | Meaning |
|---|---|---|
| bb_phir | 25.0 | Residual friction angle ϕr (°) |
| bb_jrc | 10.0 | JRC0 |
| bb_jcs | 50000.0 | JCS0 (kPa) |
| bb_l0 | 0.1 | Laboratory sample length L0 (m) |
| bb_ln | 0.1 | Field block size Ln (m) |

## CNI Rock Mass (CN)

τ = a · σc · ((σn − T0) / σc)^n

| Parameter | Default | Meaning |
|---|---|---|
| cni_sigma_c | 10000.0 | Rock mass compressive strength σc (kPa) |
| cni_a | 0.6 | Strength magnitude coefficient a |
| cni_n | 0.75 | Curvature coefficient n |
| cni_t | 100.0 | Direct tensile strength \|T0\| (kPa) |

## Generalized Hoek-Brown (HB)

σ1 = σ3 + σci (mb σ3/σci + s)^a (generalized Hoek-Brown, 2002)

| Parameter | Default | Meaning |
|---|---|---|
| sigma_ci | 25000.0 | Intact UCS σci (kPa) |
| gsi | 50.0 | GSI |
| mi | 10.0 | mi |
| d | 0.0 | Disturbance factor D |

## Hoek-Brown (mb, s, a) (HBD)

σ1 = σ3 + σci (mb σ3/σci + s)^a

| Parameter | Default | Meaning |
|---|---|---|
| sigma_ci | 25000.0 | Intact UCS σci (kPa) |
| hb_mb | 1.0 | mb |
| hb_s | 0.001 | s |
| hb_a | 0.5 | a |

### Examples

materials.py

```
project.materials = [
    MaterialSpec(name="Sand", model="MC", gamma=19, c=0, phi=34),
    MaterialSpec(name="Soft clay", model="UD", gamma=17, su=25, use_water=False),
    MaterialSpec(name="Clay", model="SH", gamma=18,
                 shansep_s=0.25, shansep_m=0.8, ocr=2),
    MaterialSpec(name="Rock", model="HB", gamma=24,
                 sigma_ci=20000, gsi=45, mi=10, d=0),
]
project.assignments = [{"point": [50, 5], "material": "Sand"}]
```

## 06 Groundwater and tension cracks

| Field | Format | Meaning |
|---|---|---|
| `water_table` | `[[x, y], ...] or None` | Piezometric line across the model. |
| `tension_crack` | `[[x, y], ...] or None` | Line whose depth below the ground defines the crack (at least 2 points, increasing x). |

### project.groundwater

| Field | Default | Meaning |
|---|---|---|
| `method` | `'water_table'` | water_table, ru or none. |
| `gamma_w` | `9.81` | Unit weight of water (kN/m3). |
| `ru` | `0.0` | Pore pressure coefficient Ru (0..1) when method = ru. |
| `crack_water` | `0.0` | Tension crack water filling, 0 (dry) .. 1 (full). |

water.py

```
project.water_table = [[0, -2], [20, -2], [45, 6], [60, 6]]
project.groundwater.method = "water_table"
project.materials[0].hu = 1.0            # per-material head factor
project.tension_crack = [[45, 7], [60, 7]]   # 3 m deep below the crest
project.groundwater.crack_water = 0.5        # half full
```

ru.py

```
project.groundwater.method = "ru"
project.groundwater.ru = 0.25
```

## 07 External loads

project.loads is a list of dictionaries.

| Key | Value | Meaning |
|---|---|---|
| `kind` | `'distributed'` | Distributed load between two points. |
| `points` | `[[x1, y1], [x2, y2]]` | Start and end on the surface (x1 ≠ x2). |
| `q` | `[q1, q2]` | Intensity at the two ends (kPa), linear in between. |
| `orientation` | `'vertical' \| 'normal'` | Load direction (default vertical). |
| `kind` | `'line'` | Point (line) load. |
| `point` | `[x, y]` | Where it acts. |
| `magnitude` | `number` | kN/m. |
| `angle` | `degrees` | Direction from +x, counter-clockwise; 270 = downward (default). |

loads.py

```
project.loads = [
    {"kind": "distributed", "points": [[45, 10], [55, 10]],
     "q": [20, 20], "orientation": "vertical"},
    {"kind": "line", "point": [50, 10], "magnitude": 60, "angle": 270},
]
```

## 08 Supports

Define support types once, then place single supports or automatic bolt patterns that refer to them by name.

| kind | Model |
|---|---|
| `end_anchored` | End anchored |
| `grouted_tieback` | Grouted tieback |
| `soil_nail` | Soil nail |
| `geotextile` | Geotextile |
| `geosynthetic` | Geosynthetic |
| `micro_pile` | Micro-pile |

## SupportType fields

| Field | Default | Meaning |
|---|---|---|
| `name` | `'Anchor 1'` | Unique name; supports and patterns refer to it. |
| `kind` | `'end_anchored'` | Support model (see the support kinds table). |
| `spacing` | `1.5` | Out-of-plane spacing (m); not used for geotextile / geosynthetic. |
| `tensile` | `200.0` | Tensile capacity: kN, or kN/m for geotextile / geosynthetic. |
| `plate` | `100.0` | Plate capacity of a soil nail (kN). |
| `bond_strength` | `50.0` | Bond strength of tiebacks and soil nails (kN/m). |
| `bond_length` | `4.0` | Bonded length of a grouted tieback (m). |
| `pullout` | `20.0` | Geotextile pull-out strength (kN/m per m). |
| `application` | `'active'` | active (force reduces driving) or passive (force divided by F). |
| `orientation` | `'parallel'` | parallel, tangent or perpendicular force direction. |
| `color` | `'#1f3d7a'` | Drawing colour. |
| `length` | `12.0` | Default length of automatically placed bolts (m). |
| `coverage` | `100.0` | Geosynthetic strip coverage (%). |
| `anchorage` | `'none'` | Geosynthetic anchorage: none, face, embedded or both. |
| `adhesion` | `0.0` | Geosynthetic interface adhesion (kPa). |
| `interface_friction` | `25.0` | Geosynthetic interface friction angle (deg). |
| `overburden_gamma` | `19.0` | Unit weight for the geosynthetic overburden stress (kN/m3). |
| `pile_shear` | `100.0` | Shear strength of one micro-pile (kN). |

## Single supports — project.supports

Each item is `{'a': [x, y], 'b': [x, y], 'type': name}`: **a** is the head on the slope face, **b** the end inside the slope.

## Bolt patterns — project.support_patterns

| Key | Meaning |
|---|---|
| `p1, p2` | [x, y] points; bolts are placed on the ground between their x values. |
| `spacing` | Distance between bolts along the slope (m, > 0). |
| `offset` | Distance of the first bolt from p1 (m, default 0). |
| `angle` | Inclination below horizontal (deg, default 15). |
| `length` | Bolt length (m, > 0). |
| `type` | Name of a SupportType. |

supports.py

```
project.support_types = [
    SupportType(name="Tieback", kind="grouted_tieback", spacing=2, tensile=300,
                bond_strength=60, bond_length=5, application="active"),
    SupportType(name="Nail", kind="soil_nail", spacing=1.5, tensile=200,
                plate=80, bond_strength=40, application="passive", length=9),
]
project.supports = [{"a": [30, 5], "b": [42, 1], "type": "Tieback"}]
project.support_patterns = [
    {"p1": [20, 0], "p2": [40, 10], "spacing": 2, "offset": 1,
     "angle": 15, "length": 9, "type": "Nail"},
]
```

geosynthetic.py

```
project.support_types = [
    SupportType(name="Geogrid", kind="geosynthetic", tensile=80, coverage=100,
                anchorage="embedded", interface_friction=28),
    SupportType(name="Pile", kind="micro_pile", spacing=1.2, pile_shear=150),
]
project.supports = [
    {"a": [22, 1], "b": [55, 1], "type": "Geogrid"},
    {"a": [35, 7.5], "b": [35, -6], "type": "Pile"},
]
```

## 09 Facings

Shotcrete with welded wire mesh on the slope face.

| Field | Default | Meaning |
|---|---|---|
| `name` | `'Shotcrete'` | Unique name; facings refer to it. |
| `thickness` | `0.1` | Shotcrete thickness (m). |
| `gamma` | `24.0` | Shotcrete unit weight (kN/m3). |
| `tau` | `400.0` | Shotcrete shear strength (kPa). |
| `mesh` | `'Q188'` | Mesh designation, e.g. Q188. |
| `mesh_as` | `188.0` | Mesh steel area (mm2/m). |
| `mesh_fy` | `500.0` | Mesh yield strength (MPa). |
| `application` | `'passive'` | passive (T/F) or active. |
| `color` | `'#7f8c8d'` | Drawing colour. |

Capacity per metre of slope = τ · thickness + As · fy. Place a facing with `{'x1': x, 'x2': x, 'type': name, 'points': [[x1, y1], [x2, y2]]}` in **project.facings**.

facing.py

```
project.facing_types = [FacingType(name="Shotcrete 10 cm", thickness=0.10,
                                   tau=400, mesh="Q188", mesh_as=188, mesh_fy=500)]
project.facings = [{"x1": 20, "x2": 40, "type": "Shotcrete 10 cm",
                    "points": [[20, 0], [40, 10]]}]
```

## 10 Seismic loading

| Field | Default | Meaning |
|---|---|---|
| kh | 0.0 | Horizontal seismic coefficient (acts in the failure direction). |
| kv | 0.0 | Vertical coefficient; positive is upward: load (1 - kv) W. |
| both_vertical | True | Also solve -kv and keep the lower factor of safety. |

**Sign of kv**
Positive kv is upward, so the vertical load is (1 − kv)·W. In Slide2 a positive kv acts downward. With both_vertical = True both signs are solved and the lower factor of safety is kept.

seismic.py

```
project.seismic.kh = 0.15
project.seismic.kv = 0.05
```

## 11 Analysis methods and settings

| Key | Method | Equilibrium |
|---|---|---|
| bishop | Bishop simplified | moment |
| corps1 | Corps of Engineers #1 | force |
| corps2 | Corps of Engineers #2 | force |
| gle | GLE/Morgenstern-Price | force and moment |
| janbu_simplified | Janbu simplified | force |
| janbu_corrected | Janbu corrected | force (corrected) |
| lowe_karafiath | Lowe-Karafiath | force |
| ordinary | Ordinary/Fellenius | moment |
| spencer | Spencer | force and moment |

## project.methods

| Field | Default | Meaning |
|---|---|---|
| `methods` | `[…]` | Limit equilibrium methods to run (keys in the methods table). |
| `n_slices` | `50` | Number of slices (at least 3). |
| `tolerance` | `0.005` | Convergence tolerance on the factor of safety. |
| `max_iter` | `75` | Maximum solver iterations. |
| `interslice` | `'half-sine'` | GLE interslice function: half-sine, constant, clipped-sine, trapezoidal. |
| `f_start` | `0.5` | Trapezoidal / clipped function value at the start (0..1). |
| `f_end` | `0.5` | Trapezoidal / clipped function value at the end (0..1). |

## project.general

| Field | Default | Meaning |
|---|---|---|
| `units` | `'metric'` | Unit system; metric (m, kN, kPa). |
| `failure_direction` | `'right_to_left'` | auto, right_to_left or left_to_right. |
| `data_output` | `'standard'` | Amount of result data kept: standard. |

## project.advanced

| Field | Default | Meaning |
|---|---|---|
| `check_admissibility` | `True` | Reject surfaces with inadmissible slice forces. |
| `min_depth` | `0.0` | Minimum slide depth (m). |
| `m_alpha_min` | `0.2` | Admissibility threshold for m-alpha (0..1]. |
| `e_max_ratio` | `50.0` | Reject interslice forces above ratio x total weight. |

methods.py

```
project.methods.methods = ["bishop", "spencer", "gle"]
project.methods.n_slices = 40
project.methods.interslice = "half-sine"
project.general.failure_direction = "right_to_left"
project.advanced.min_depth = 1.0
```

# 12 Slip surfaces

project.surfaces selects the surface type and the search.

| Field | Default | Meaning |
|---|---|---|
| surface_type | 'circular' | circular, noncircular or user. |
| search | 'auto_refine' | Search strategy: auto_refine. |
| effort | 'normal' | Search effort: fast, normal, fine. |
| n_ctrl | 4 | Interior vertices of non-circular surfaces. |
| noncircular_shape | 'polyline' | polyline (straight segments) or spline. |
| convex_only | True | Keep non-circular surfaces convex. |
| seed | 0 | Random seed of the search (reproducible results). |
| left_range | None | Slope limits for the left end: [x_min, x_max] or None. |
| right_range | None | Slope limits for the right end: [x_min, x_max] or None. |
| user_surface | None | User polyline [[x, y], ...] when surface_type = user. |
| user_circle | None | Exact fixed circle [xc, yc, radius] when surface_type = user. |

surfaces.py

```
# automatic non-circular search with slope limits
project.surfaces.surface_type = "noncircular"
project.surfaces.effort = "normal"
project.surfaces.left_range = [10, 25]
project.surfaces.right_range = [40, 60]
```

fixed_surface.py

```
# analyse one given circle only
project.surfaces.surface_type = "user"
project.surfaces.user_circle = [25, 30, 30]   # xc, yc, radius
```

# 13 Notes and display options

## Notes on the drawing — project.annotations

| Key | Meaning |
|---|---|
| `kind` | text, line, arrow, rect or ellipse. |
| `pts` | [[x, y]] for text; [[x1, y1], [x2, y2]] for the other shapes. |
| `text` | Text content (text notes); '\n' starts a new line. |
| `color` | Line or text colour, e.g. '#c0392b'. |
| `width` | Line width. |
| `fill` | Fill colour of rect / ellipse, or None. |
| `size, bold, box` | Text size, bold text and frame around the text. |

notes.py

```
project.annotations = [
    {"kind": "text", "pts": [[30, 16]], "text": "Design section A-A",
     "color": "#1b2a3a", "size": 11, "bold": True, "box": True},
    {"kind": "arrow", "pts": [[30, 15], [26, 3]], "color": "#c0392b", "width": 1.5},
]
```

## project.display

| Key | Default | Meaning |
|---|---|---|
| external_color | '#606060' | External boundary colour. |
| material_boundary_color | '#808080' | Material boundary colour. |
| water_table_color | '#1f5fd1' | Water table colour. |
| show_vertices | True | Draw boundary vertices. |
| line_width | 2 | Boundary line width. |
| show_ruler | True | Show the rulers. |
| show_grid | True | Show the grid. |
| fill_materials | True | Fill regions with material colours. |
| show_slope_limits | True | Show the slope limit markers. |
| show_slices | False | Draw the slices of the critical surface. |
| slice_fill_color | '#f6d65a' | Slice fill colour. |
| show_lines_to_center | True | Lines from the critical circle to its centre. |
| show_fs_text | True | Write the factor of safety at the centre. |
| show_all_surfaces | False | Draw the lowest surfaces, not only the critical one. |
| show_fs_contours | True | Slip-centre FS contour map (circular searches). |
| critical_dashed | True | Critical surface as a dashed line. |
| show_legend | True | Material legend. |
| show_fs_legend | True | FS legend of all methods. |
| material_labels | 'none' | Region labels: none, name or full. |
| label_size | 8 | Label font size. |
| show_control_points | False | Control polygon of a spline critical surface. |

display.py

```
project.display["material_labels"] = "full"
project.display["show_slices"] = True
```

# 14 A complete model

Everything together: copy, run, then Compute.

complete_model.py

```
from slipsurface.project import empty_project, MaterialSpec
from slipsurface.loads import SupportType, FacingType

project = empty_project()
project.summary.title = "Scripted layered cut"
project.external = [[0, -8], [48, -8], [48, 12], [26, 12], [20, 0], [0, 0]]
project.materials = [
    MaterialSpec(name="Soil", model="MC", gamma=19, c=8, phi=30),
    MaterialSpec(name="Rock", model="HB", gamma=23, sigma_ci=8000, gsi=40, mi=8),
]
project.material_boundaries = [[[0, -3], [48, -3]]]
project.assignments = [
    {"point": [24, 5], "material": "Soil"},
    {"point": [24, -6], "material": "Rock"},
]
project.water_table = [[0, -2], [20, -2], [40, 4], [48, 4]]
project.tension_crack = [[30, 10], [48, 10]]
project.groundwater.crack_water = 0.5
project.support_types = [
    SupportType(name="Nail", kind="soil_nail", spacing=1.5, tensile=240,
                plate=80, bond_strength=40, application="passive", length=9),
]
project.support_patterns = [
    {"p1": [20, 0], "p2": [26, 12], "spacing": 2, "offset": 1,
     "angle": 15, "length": 9, "type": "Nail"},
]
project.facing_types = [FacingType(name="Shotcrete")]
project.facings = [{"x1": 20, "x2": 26, "type": "Shotcrete",
                    "points": [[20, 0], [26, 12]]}]
project.loads = [
    {"kind": "distributed", "points": [[29, 12], [40, 12]],
     "q": [15, 15], "orientation": "vertical"},
    {"kind": "line", "point": [42, 12], "magnitude": 20, "angle": 270},
]
project.seismic.kh = 0.1
project.methods.methods = ["bishop", "spencer"]
project.methods.n_slices = 40
project.surfaces.effort = "fast"
print("Ready:", project.summary.title)
```

# 15 Troubleshooting

| Message | Cause and fix |
|---|---|
| `Define at least one material.` | project.materials is empty. |
| `Material names must be unique.` | Two materials share a name. |
| `Assigned material 'X' is not defined.` | An assignment or the material list has a typo. |
| `Support type 'X' is not defined.` | A support or pattern names a missing SupportType. |
| `Unknown method: X` | Use a key from the methods table. |
| `unknown option 'X'` | A choice field has a value that is not allowed; see the tables. |
| `No external boundary defined` | project.external needs at least 3 points. |
| `project must be a Project instance` | The name project was reassigned to something else. |

**Debugging**
Use print() freely, and call project.validate() inside the script to see a validation error at the line where you made the change.